



Sentinelle Vanguard VG Pre-Installation Manual

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Revision History

Revision	Date	Author	Type (Major/Minor)	Summary of Changes
1.0	22 OCT 2007	Dylan Laister	Major	Initial Release.
1.1	11 JAN 2008	Dylan Laister	Minor	Editing of content
2.0	15 FEB 2008	Dylan Laister	Major	Reformatted and added figures
3.0	4 MAR 2008	Dylan Laister	Major	Added figures and descriptions; Minor clarifications to text

Table of Contents

1 Purpose.....	3
2 Scope	3
3 Responsibility	3
4 Definitions.....	3
5 Preparing for Installation.....	3
6 GE Signa and Sentinelle Vanguard Interface.....	4
Table 2: Vanguard / Signa Interface Specifications	5
7.0 Figure reference from Table 2: Vanguard/Signa Interface Specifications.....	6

1 Purpose

To define the necessary and sufficient site conditions to permit successful installation of the Sentinelle Vanguard Breast Auxiliary Table.

2 Scope

2.1 The interface requirements are solely intended to define the interface of the Vanguard to any GE 1.5T MRI

3 Responsibility

3.1 It is the responsibility of Sentinelle Medical and GE Healthcare to ensure that the interface dimensions specified in this document remain static and that no changes to the equipment affecting these dimensions are undertaken as this will impact the installation and function of the Vanguard. If it is necessary to adjust the scanner's dock, bridge, interlocks, switches or other siting conditions, Sentinelle Medical must be notified so that in-service can be planned.

4 Definitions

- 4.1 SMI – Sentinelle Medical Inc.
- 4.2 Stretcher – (or Vanguard Stretcher) the lower portion of the Vanguard which rolls on casters and carries the patient support. The GEHC equivalent term for this part is “table”
- 4.3 Patient Support – (or Vanguard Breast MRI Auxiliary Tabletop) the upper portion of the Vanguard which supports the patient and rolls in and out of the scanner bore. The GEHC term for this part is “cradle”

5 Preparing for Installation

5.1 In order to successfully install the Vanguard, the following pre-installation details in table 1 below must be verified before the sale of the Vanguard and in advance of shipping of the Vanguard:

Table 1

Check	Description
	Site contact information has been acquired and entered into SMI records.
	Site staff has agreed to delivery and installation dates/times and have arranged site access for SMI staff with security, engineering, charge tech and any other necessary personnel.
	Site has adequate dock facilities to receive crated Vanguard and elevator to transport it to scanner.
	Check to ensure no doors on path to MRI are narrower than 30” (760mm).
	Site has identified the particular scanner which is to have the system installed, and that system is up and running.
	Configuration of system has been identified (select one of): O Excite (software rev 11.0) (legacy) O HD (software rev 12.0) (legacy/hypertronix)

	O HDx (software rev 14.0) (legacy/hypertronix) O Forward HDx (14.0) (hypertronix connector only)
	Site has adequate indoor storage space for the crate and staff has been instructed not to open crate.
	Latest Service Patch has been identified in order to confirm presence of Vanguard coil config file.
	Site has 760mm x 2160mm (30" X 85") storage space for the Vanguard in the scanner room or other accessible space after installation.
	Site has concurred to requirement of shipping crate/packaging disposal or return shipment of crate.

6 GE Signa and Sentinelle Vanguard Interface

6.1 Interface

The Vanguard system is a receive-only coil array integrated into a dedicated cradle and table. Various coil elements may be interchanged to reconfigure the array. All of the coils are receive-only. Although the system docks to the scanner in essentially the same way as the Signa table, the system does not use the up/down controls provided on the scanner. The Vanguard stretcher incorporates various adjustments to accommodate a range of site and scanner conditions. It is not required to adjust any part of the Signa MRI or table in order to install or use the Vanguard, provided the site conditions are within the bounds outlined below (see Table 1).

When docked to the GE Signa Scanner, the Vanguard's patient support (cradle) is pulled to the isocenter by the Carriage. The stretcher (table) can be adjusted to accommodate normal variations in the geometry of the scanner as outlined in this document. If the bridge, body coil, dock or related features of the scanner are adjusted, a Sentinelle service call is required for re-alignment of the Vanguard. All defined interface dimensions and tolerance ranges are listed in Table 2.

If the scanner is being updated to a Coil Config revision or service patch which does not incorporate the Vanguard, an SMI in-service visit is required to re-enter the coil config's manually. Sentinelle Medical's Service department shall be notified if any such changes are planned or undertaken.

Table 2: Vanguard / Signa Interface Specifications

ID	Description	Figure Reference	Nominal	Limits	Notes
1	Bore Diameter, through entire bore	Fig 1	600 mm	595mm min, no max	It is acceptable to compress Velcro with light finger pressure while measuring. The areas of the bore's light housings are excluded from this measurement.
2	Dock Face squareness to bore.	Fig 2	0 (square)	20mm max	Measured between Bridge Guide Rail/Rubber Bumpers (see Fig 5)
3	Dock Face position (Head-Foot direction)	Fig 1	300mm	+/- 25mm	Measured between dock face and face of bridge.
4	Dock Face position from center (Left/right direction)	Fig 3	0	+/- 15mm	Measured from centre line of bridge to centre line of dock.
5	Bridge Centered in Body Coil/Bore (Left/Right)	Fig 1	0	+/- 2.5mm	Using Bridge Guide Rail for reference
6	Bridge level to floor in Head-Foot direction	N/A	0	40mm max	
7	Bridge twist	N/A	0	15mm total	Measured with long straightedge on bridge surface, over entire length
8	Isocenter to floor distance	N/A	1070mm	+/- 3mm	
9	Bridge to floor distance	N/A	885.7mm	+/- 6mm	
10	Scanner Room Floor Flatness (total variation in a 1m ² area)	N/A	0	3mm	Larger variations are permissible but affect perceived quality of device due to rocking.
11	Independent cradle latch adjuster on Carriage ¹	Fig 4	12mm	+/-6mm	
12	Advance to scan force/holding force	N/A	12.9kgf	+/- 0.7kgf	
13	Protrusion Dock switch rod on face of Dock	Fig 5	10mm	+/-5 mm	As measured in open and closed positions
14	Cradle latch motion is unimpeded	Fig 4	--	--	This is necessary for effective operation of the Emergency Egress.
15	Scanner has functioning Emergency Undock mechanism	N/A	--	--	
16	Maximum discontinuity of floor upon entering scan room	N/A	0 (no lip)	25mm max	Grade over transition must be < 25%

7.0 Figure reference from Table 2: Vanguard/Signa Interface Specifications

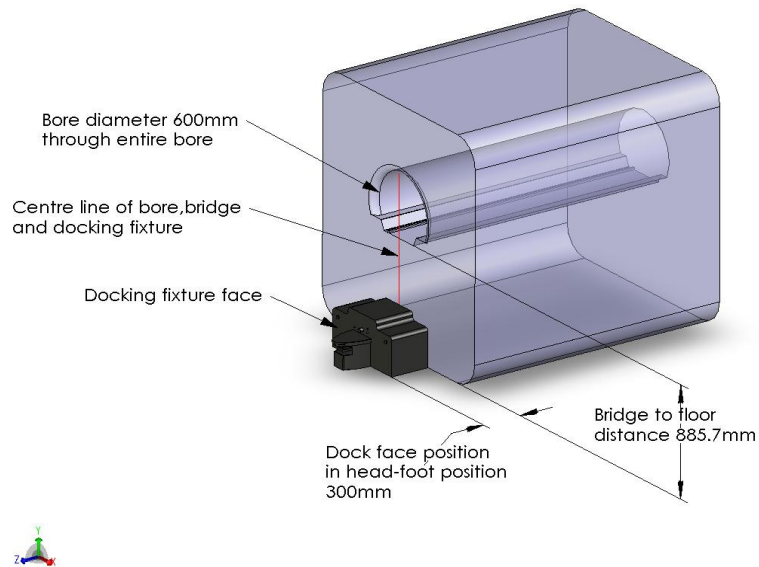


Figure 1: Scanner and Dock

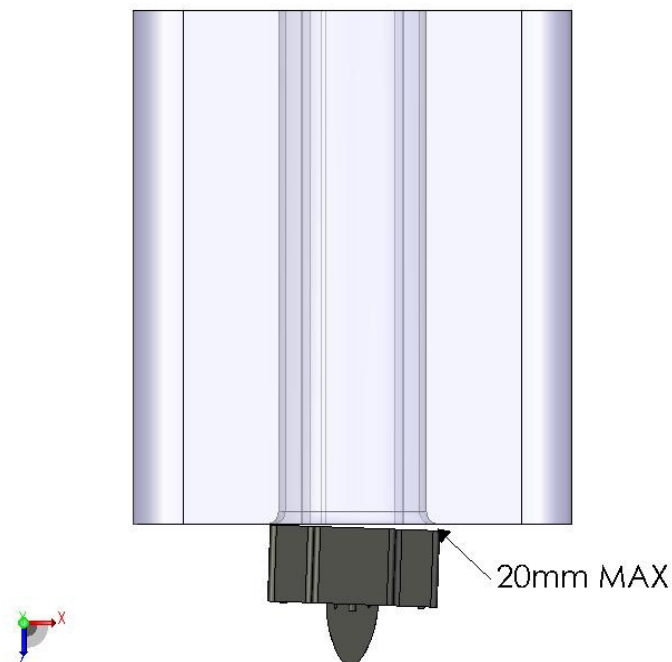


Figure 2: Dock Angle

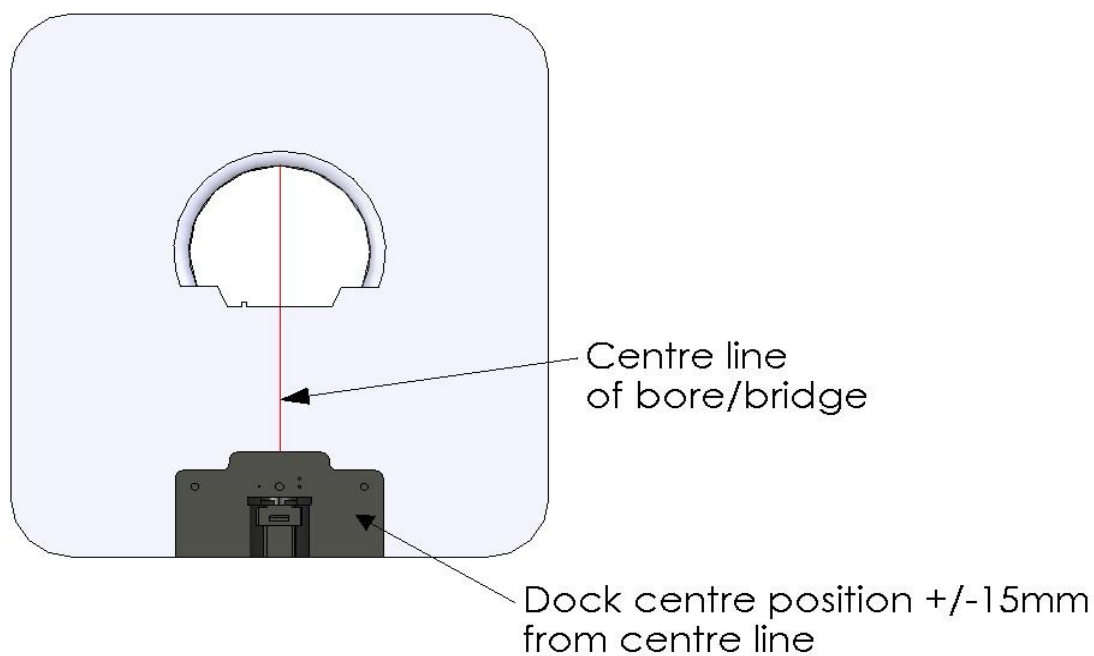


Figure 3 – Dock centering

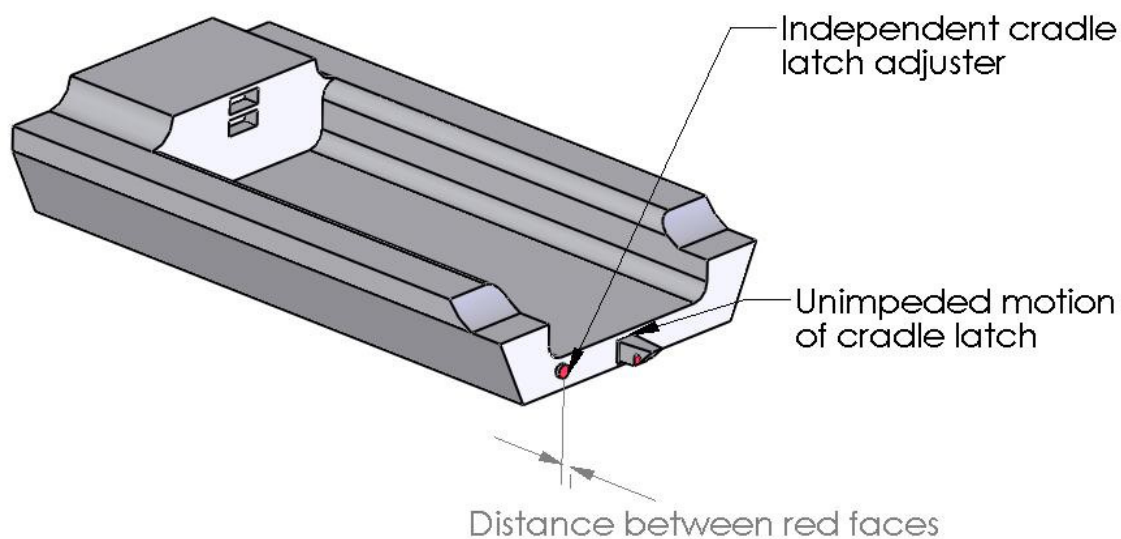


Figure 4 - Independent cradle latch adjuster

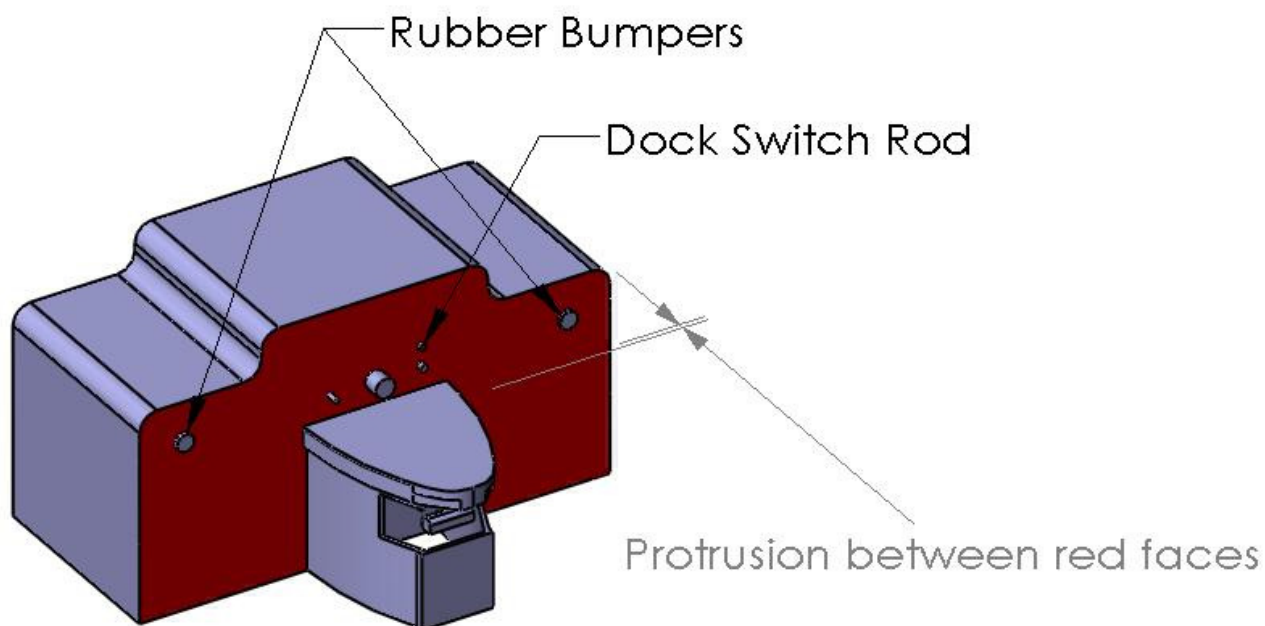


Figure 5 - Dock switch protrusion

Notes

¹ The independent cradle latch is located at the foot end of the table. An adjustable plunger protrudes from the cradle and is depressed after docking when the carriage advances out of the MRI's bore. It is sometimes referred to as "doghouse interlock"